

Asset Pricing and Valuation

Lecture 7: Options and Binomial Trees

Matthew Smith

Options pricing

look at

<https://financetrain.com/binomial-option-pricing-model-in-r>

Binomial trees

- Used for pricing options
 - A stock price can follow many paths
 - Difficult to know with accuracy the path as T increases.
- Construct a binomial tree representing different possible paths that might be followed by the stock price over the life of an option
- Stock price follows a **random walk**
- process by which randomly-moving objects wander away from where they started
 - It has a probability of moving up or down a certain **percentage** point



Stock price paths simulation

Overview

- Objective: Want to price an option
- Based on the concept of no-arbitrage
- **Binomial option pricing models** is a risk-free method for estimating the value of path-dependent alternatives.
- We can determine how likely we are to **buy** or **sell** at a given price in the future.
- According to this model: The current option value is equal to the **present value** of the probability weighted **future** payoffs of the investment.

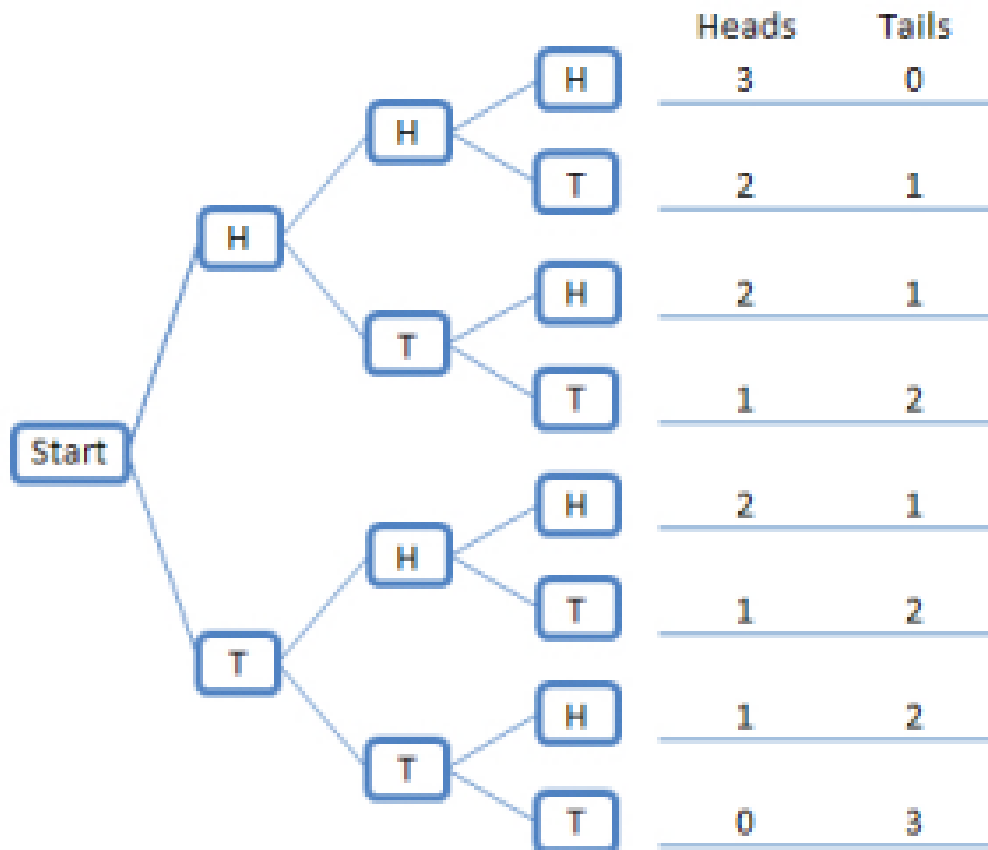
Basics of the Binomial Option Pricing Model

- As an investor, we are aware of the current stock price at any time.
- We want to predict future changes in stock prices
- We divide the time until option expiry into **equal** parts under this scenario (t = weeks, months, quarters)
- The binomial model uses an iterative process for each period to determine how likely the movement will be (up or down)
- We create a binomial distribution of stock prices.

Binomial distribution (model)

Binomial distribution (model)

- A binomial sequence is characterized by a series of Yes/No occurrences
- Consider a series of coin tosses (yes/no) - only two results (binomial)
- Trials (periods) are said to be independent of each other.
- Consider the following: Possible paths (coin toss) for 3 periods - each period has 50/50 probability



Binomial distribution (model)

Binomial distribution (model)

- In stock investing, the binomial model is used for stock prices (up or down)
 - Period examined is broken into intervals - months, weeks, days, hours, minutes
 - At each point evaluated point the stock must therefore satisfy

$$S_u = S(1 + u)$$

or

$$S_d = S(1 + d)$$

- Where S_u is stock up, S_d is stock down, u is percentage increase and d is percentage decrease

Binomial distribution (model)

- The outcome of the up and down movements produces a binomial stock tree
- The tree describes all potential price paths that the **underlying** can take until expiry

American Call Option

Start	Trials			Payoff
	1	2	3	
			115	15
		110		
	105		105	5
100		100		
	95		95	0
		90		
			85	0

Binomial options pricing model

- To determine the accurate pricing for an asset is difficult
 - Which is why stock pricing change all the time (up/down)
 - Companies don't change their valuation on a daily basis -but their stock pricing fluctuate every second.
 - Question: What is the right current price today for an expected future payoff?
- Binomial options pricing model:
 - Values options using an *iterative* approach utilizing multiple periods to value American options.
 - Two possible outcomes in each iteration - **up** or **down**
 - From each iteration we obtain a **binomial tree**
 - Used often in practice (more than Black-Scholes model)

Binomial options pricing model

Binomial options pricing model

- We have 2 assets which the valuation depends on
 - the **call option**
 - the **underlying stock**
- Both parties agree the stock price will move from \$100 in one year (no other price moves possible)
 - A to \$110
 - B to \$90

Binomial options pricing model

- We create a portfolio of the two assets
 - A call option
 - Underlying stock
- **objective** - create a portfolio such that regardless of where the underlying price goes \$110 or \$90 the return on the portfolio is **always** the same.
- Suppose we buy d shares of the underlying and **short one call** to create this portfolio
 - If the price goes to \$110 our shares are worth $\$110 * d$ and we lose \$10 on the **short call** payoff
 - Portfolio net value is $(110d - 10)$ If the price goes to \$90 our shares are worth $90 * d$ (option expires worthless)
 - Portfolio net value is $90 * d$
- We want the portfolio value to be the same regardless of the underlying stock price movement
 - Therefore we want $110d - 10 = 90d$
 - Therefore $d = 1/2$ (half a share of underlying)
 - Portfolio value is \$45 in one years time
- Need to compute the present value (discounting it by the risk-free rate)

$$\begin{aligned}\text{Present value} &= 90d \times e^{-5\% \times 1\text{Year}} \\ &= 45 \times 0.9523 \\ &= 42.85\end{aligned}$$

Binomial options pricing model

- At present, the portfolio has $1/2$ a share of the underlying (priced at \$100 and one **short call**)
- Portfolio should equal the present value

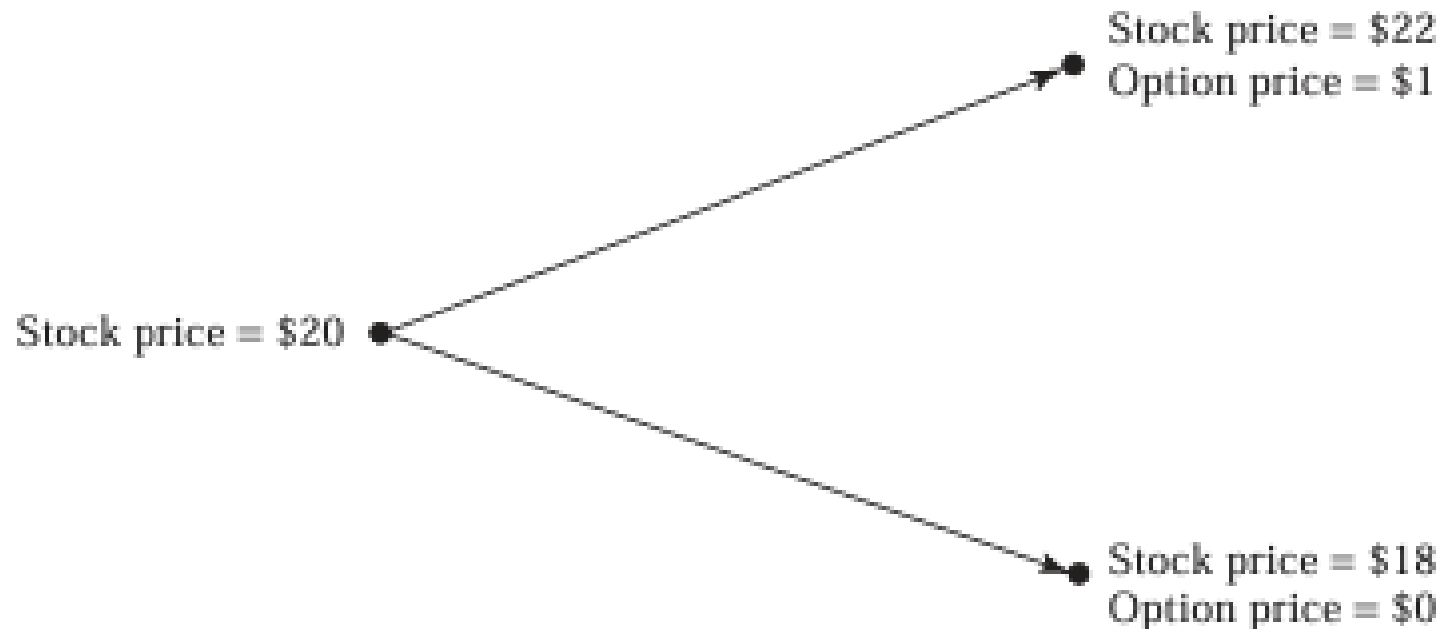
$$\frac{1}{2} \times 100 - 1 \times \text{Call price} = \$42.85$$

- Assuming that the portfolio value remains the same regardless of the underlying stock price fluctuations (risk free of the underlying movements)
- Our portfolio is neutral to price movement risk and earns the risk-free rate of return
- Both trader A and B are willing to pay the same \$7.14 for this call option (despite their different perception of probabilities 60% and 40%)
- The call option price is worth \$7.14 today
- Where is the **volatility** in the pricing of the option?
 - The volatility is included in the problems definition
 - Assuming 2 (binomial) states of price levels (\$110 and \$90) - volatility is implicit in the assumption of the two price levels

One step binomial model

One step binomial model

- Consider a stock is trading at \$20
- We assume the stock could be worth \$22 or \$18 in 3 months
- We want to value a European **call** option to buy the stock for \$21 in 3 months
- The option will have one of two values in 3 months
 - If stock price is \$22 then the option is valued at \$1
 - If stock price is \$18 then the option is valued at \$0



One step binomial model

- Definitions:
 - A European **call** option gives the holder the **right** but **not the obligation** to **buy** an asset at a **specified price** on a **specified date**
 - Therefore: **Writing a call**: We **sell** somebody (buyer of the call) the **right** to purchase an asset from us (writers of the call) for a **specified price** on a **specified date**.
- Consider a portfolio consisting of;
 - A long position in Δ (Delta) shares of a stock (own the shares)
 - A short position in one call option (sell the option for somebody to buy X shares)
- We want to calculate the value of Δ which makes the portfolio riskless
- Case 1
 - If stock price moves up from \$20 to \$22 then the value of shares is 22Δ ($22 * \# \text{ shares}$) and the value of the option is 1
 - The total value of the portfolio is $22\Delta - 1$
- Case 2
 - Stock goes down \$20 to \$18 then the value of shares is 18Δ and value of option is 0
 - Total value of the portfolio is 18Δ

One step binomial model

Set the portfolios equal to each other

$$\begin{aligned}22\Delta - 1 &= 18\Delta \\ \Delta &= 0.25\end{aligned}$$

- A riskless portfolio is therefore;
 - Long 0.25 shares
 - Short 1 option
- Stock price to \$22 then the portfolio is worth $22 \times 0.25 - 1 = \$4.5$
- Stock price to \$18 then the portfolio is worth $18 \times 0.25 - 0 = \$4.5$
- Regardless of stock price fluctuations the value is always \$4.5 (at option expiry)

One step binomial model

- A riskless portfolio must earn the risk free rate of interest
- Suppose the risk-free rate is 12%
- If a riskless portfolio is worth \$4.5 in 3 months time - we need to discount it back to the present value e^{-rT}

$$\begin{aligned} &\text{portfolio value today} \\ &4.5 \times e^{-0.12 \times 3/12} = 4.367 \end{aligned}$$

Therefore, if the value of the stock today is known to be \$20

- Denote the option price to be f the value of the portfolio is:

$$\begin{aligned} 20\Delta - f &= \text{portfolio value} \\ 20 \times 0.25 - f &= \text{portfolio value} \end{aligned}$$

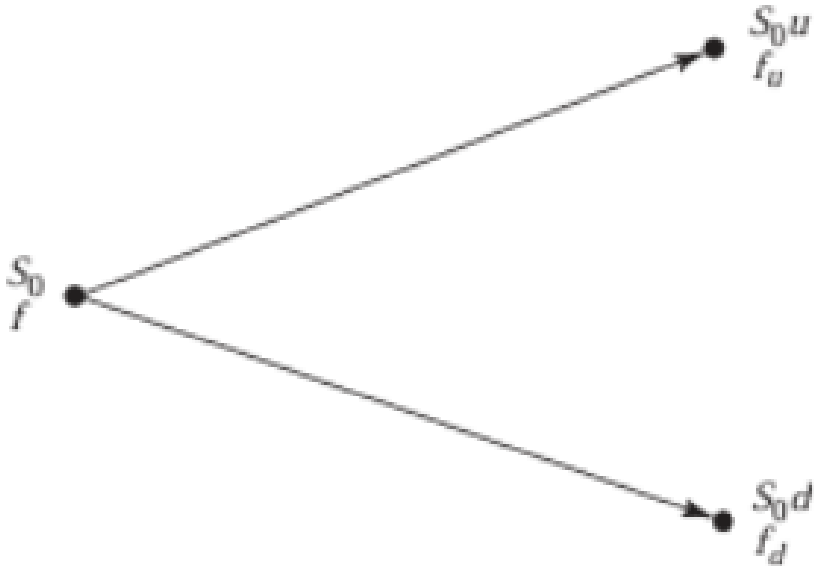
- We know the riskless portfolio is worth 4.367, so

$$\begin{aligned} 5 - f &= 4.367 \\ f &= 0.633 \end{aligned}$$

- Therefore the current value of the option must be 0.633
- If $f > 0.633$ the portfolio would cost less than 4.367 (earning more than the risk free rate)
- If $f < 0.633$ shorting the portfolio, we could borrow money at less than the risk free rate.

Generalization - One step model

- Consider a stock trading at S_0 and a stock option f , with expiry at T
- Stock can move **up** to S_0u ($u > 1$) or **down** to S_0d ($d < 1$) in one period
- The percentage increase in an upwards stock movement is $u - 1$
- The percentage decrease in a downwards stock movement is $1 - d$
- If stock moves up to S_0u the payoff from the option f_u
- If stock moves down to S_0d the payoff from the option f_d



Generalization - One step model

Generalization - One step model

- For a riskless portfolio it must earn the risk free rate of interest
- Suppose the risk-free rate is r
- If a riskless portfolio worth $S_0 u \Delta - f_u$ in T months time (future value) - we need to discount it back to the present value using e^{-rT}

$$(S_0 u \Delta - f_u) e^{-rT}$$

- So r is our risk free rate - then $(S_0 u \Delta - f_u) e^{-rT}$ is our riskless portfolio in PV terms.

- **NOTE: Similar to previous answer where we found our riskless portfolio value of \$4.5 (future value) and had to discount it back to PV $4.5 \times e^{-0.12 \times 3/12} = 4.367$ using the risk free rate of 12%**

$$S_0 \Delta - f = \text{portfolio value today}$$

- **NOTE: Similar to previous example $20\Delta - f$**
- We know the riskless portfolio **today** is worth $(S_0 u \Delta - f_u) e^{-rT}$, so our portfolio value **today** equals the risk free portfolio value

$$S_0 \Delta - f = (S_0 u \Delta - f_u) e^{-rT}$$

- **NOTE: Previously our PV of the riskless portfolio was worth 4.367, we plug in the same values Similar to our $5 - f = 4.367$**
- Solving for f to get our options price - of a risk-free interest portfolio

$$f = S_0 \Delta (1 - u e^{-rT}) + f_u e^{-rT}$$

Generalization - One step model

- We know that;
- Δ - ratio of the change in the option price to the change in the stock price
- f - current value of the options price

$$\Delta = \frac{f_u - f_d}{S_0 u - S_0 d}$$
$$f = S_0 \Delta (1 - u e^{-rT}) + f_u e^{-rT}$$

- Substituting Δ into f

$$f = S_0 \left(\frac{f_u - f_d}{S_0 u - S_0 d} \right) (1 - u e^{-rT}) + f_u e^{-rT}$$

- or;

$$f = \frac{f_u (1 - d e^{-rT}) + f_d (u e^{-rT} - 1)}{u - d}$$

- or;

$$f = e^{-rT} [p f_u + (1 - p) f_d]$$
$$\text{Where; } p = \frac{e^{rT} - d}{u - d}$$

Generalization - One step model

We have:

$$f = e^{-rT} [pf_u + (1 + p)f_d]$$

$$\text{Where; } p = \frac{e^{rT} - d}{u - d}$$

Generalization - One step model

- Consider a stock trading at S_0 and a stock option f , with expiry at T
- Stock can move **up** to $S_0 u$ ($u > 1$) or **down** to $S_0 d$ ($d < 1$) in one period
- The percentage increase in an upwards stock movement is $u - 1$
- The percentage decrease in a downwards stock movement is $1 - d$
- If stock moves up to $S_0 u$ the payoff from the option f_u
- If stock moves down to $S_0 d$ the payoff from the option f_d
- Consider the following example:
 - $u = 1.1, d = 0.9, r = 0.12, T = 0.25, f_u = 1$ and $f_d = 0$
- Recall,

$$p = \frac{e^{rT} - d}{u - d}$$

- So;

$$p = \frac{e^{0.12 \times 3/12} - 0.9}{1.1 - 0.9} = 0.6523$$

- Recall p is the probability of an up movement and $(1 - p)$ is a probability of a down movement.
- The options payout given by plugging p into:

$$f = e^{-rT} [p f_u + (1 - p) f_d]$$
$$f = e^{-0.12 \times 3/12} [0.6523 \times 1 + (1 - 0.6523) \times 0] = 0.633$$

- Which is exactly what we got in our previous answer.

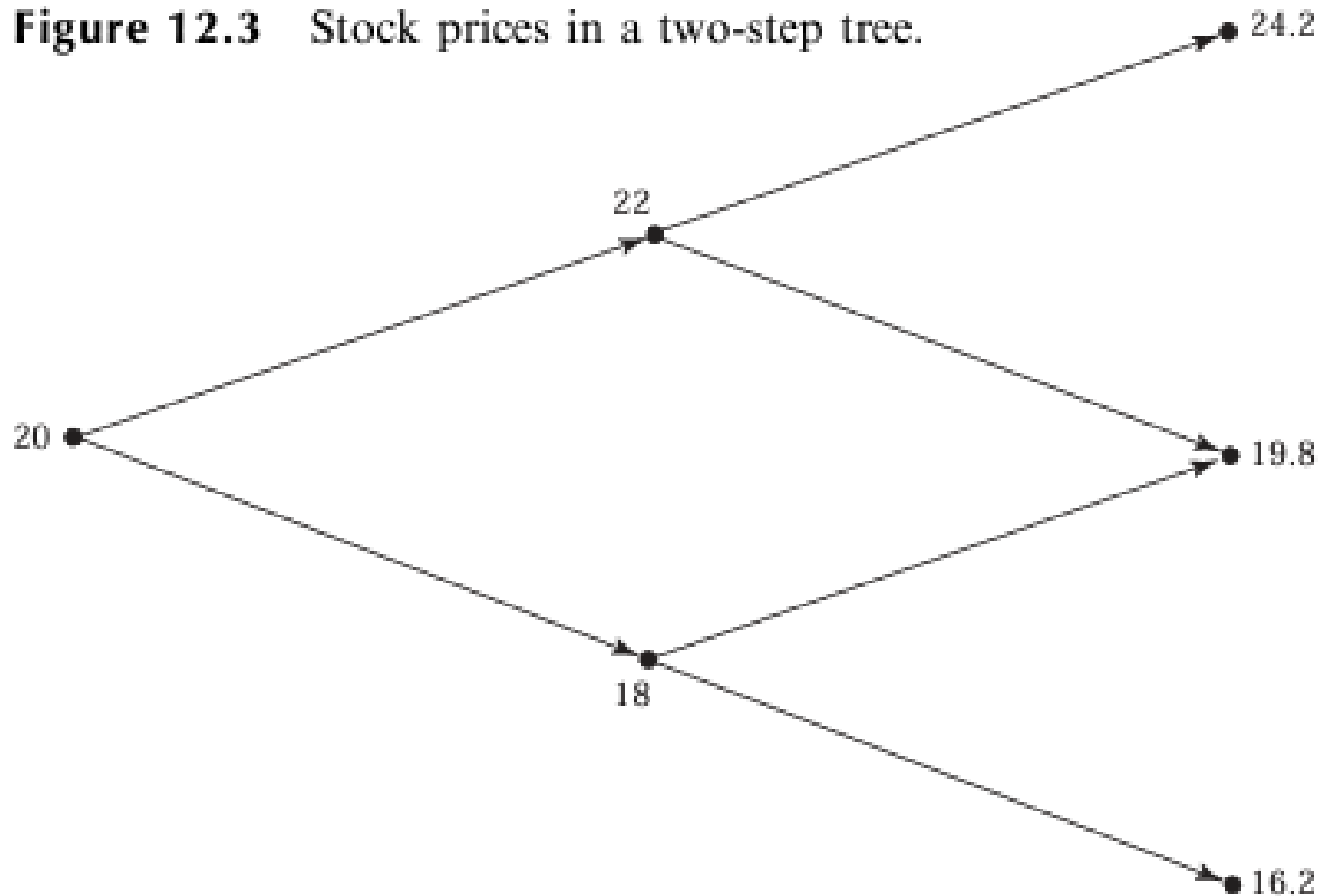
Revisit example 1

Two step binomial tree

- Stock price is \$20
- May go up by 10% or down by 10% in both periods
- Each time step is 3 months
- risk free rate $r = 12\%$ per annum
- Consider a 6 month option with a strike of \$21
- We want to calculate the options price today

Two step binomial tree

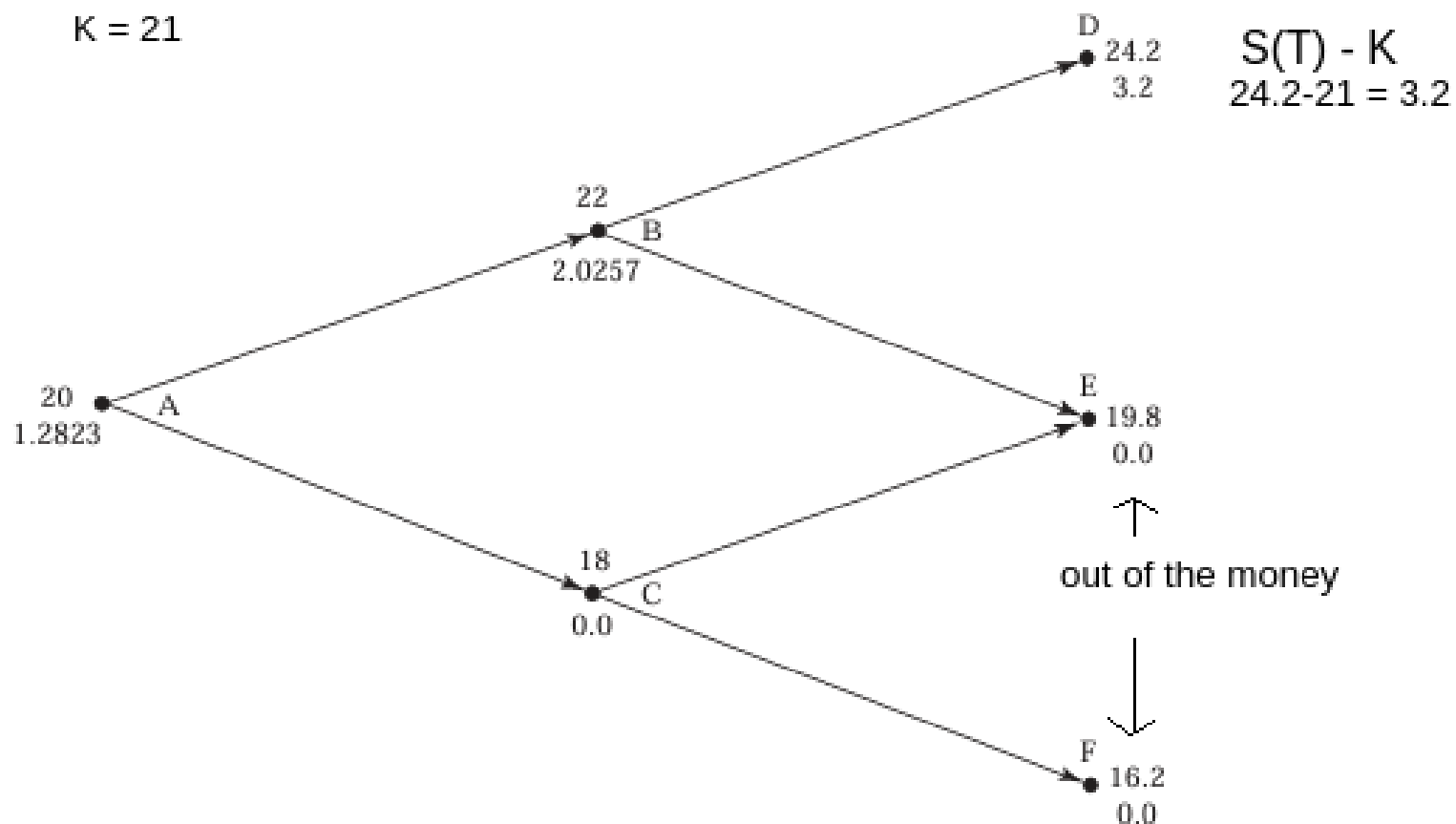
Figure 12.3 Stock prices in a two-step tree.



Two step binomial tree

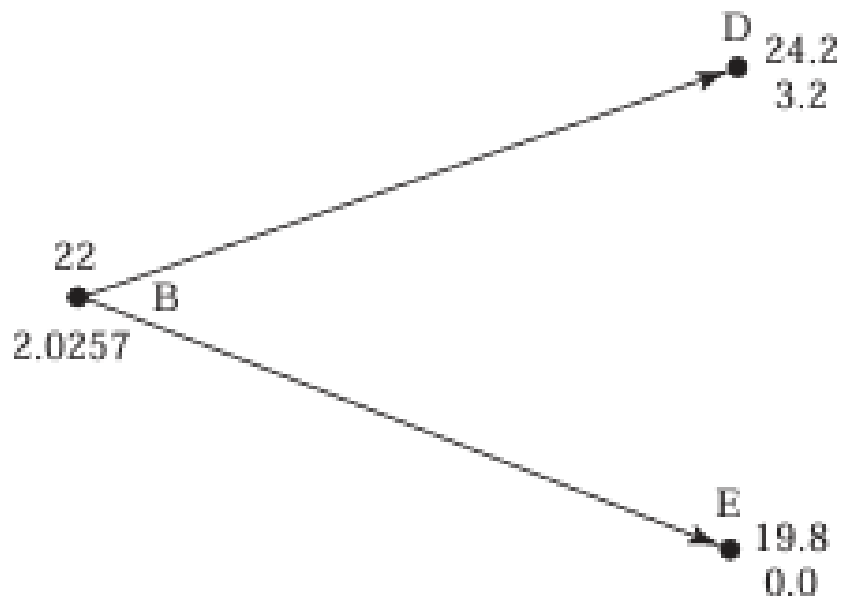
Figure 12.4 Stock and option prices in a two-step tree. The upper number at each node is the stock price and the lower number is the option price.

$K = 21$



Two step binomial tree

- Calculate the options price at node B
- Suppose $u = 1.1, d = 0.9, r = 0.12, T = 0.25$, we know $p = 0.6523$
- Recall; $f = e^{-rT}[pf_u + (1 + p)f_d]$
- $e^{-0.12 \times (3/12)}(0.6523 \times 3.2 + 0.3477 \times 0) = 2.0257$
- Now we know the options price at node B is 2.0257
- Value at node A is $e^{-0.12 \times (3/12)}(0.6523 \times 2.0257 + 0.3477 \times 0) = 1.2823$



Delta

- A greek letter and used in pricing and hedging of options
- The Δ of a stock option is the ratio of the change in the price of the stock option to the change in the price of the underlying stock.
- Its the number of units of stock we should hold for each option shorted in order to create a riskless portfolio.
- Delta hedging is the creation of a **riskless** portfolio
- Δ of a call option is **positive**
- Δ of a put option is **negative**

Delta

- Recall;

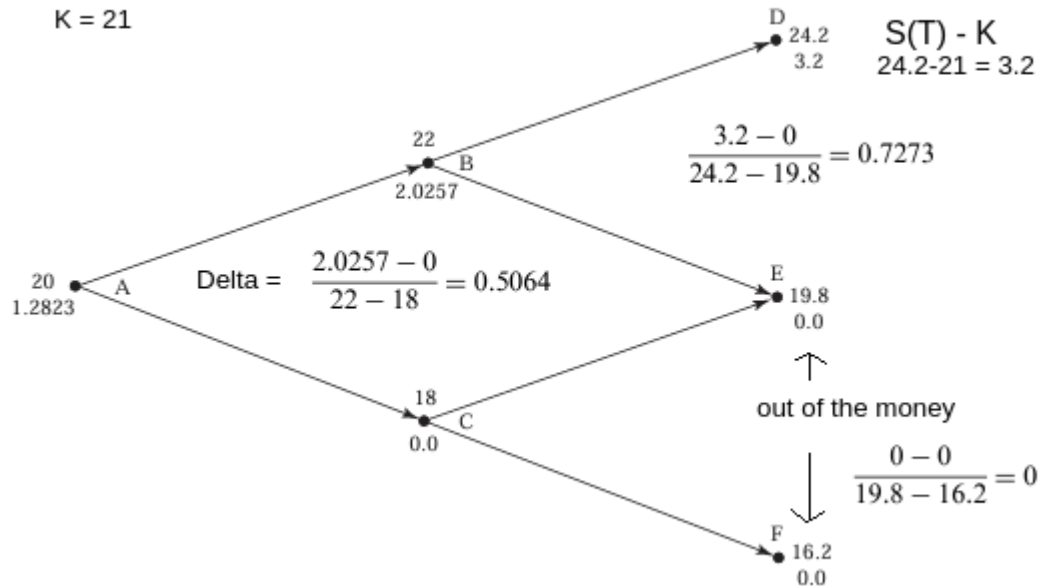
$$\frac{1 - 0}{22 - 18} = 0.25$$

- So when the stock price changes from 18 to 22, the option price changes from 0 to 1

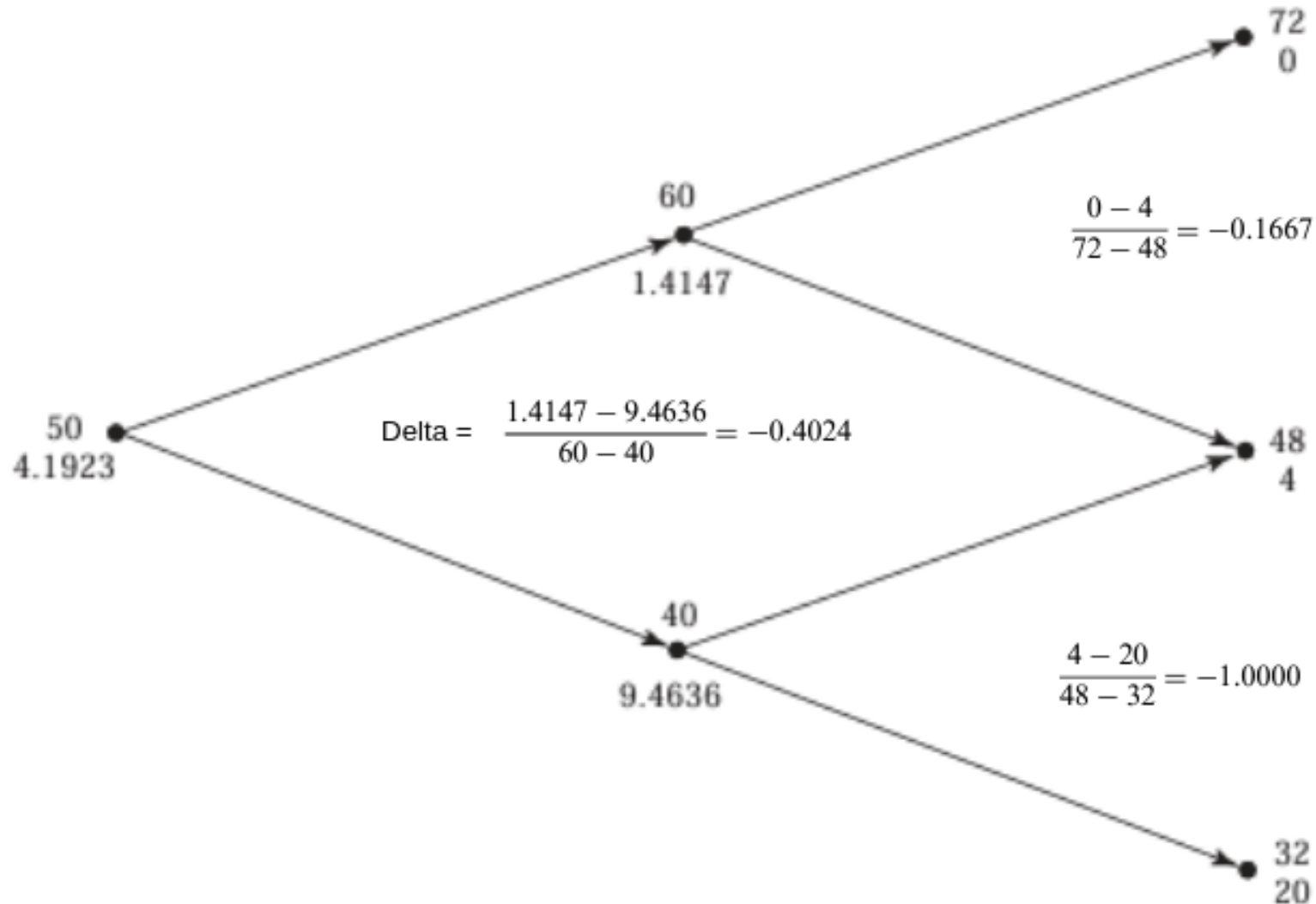
- Recall (in the 2 step binomial example)
- We can compute the Δ 's for each time step for each movement (upwards or downwards)

Figure 12.4 Stock and option prices in a two-step tree. The upper number at each node is the stock price and the lower number is the option price.

$K = 21$



Delta (Example 2)



Delta

- In both examples, Δ changes over time
- In the first, from 0.5064 to either 0.7273 or 0
- In the second, from -0.4024 to either -0.1667 or - 1.000

